

Disc Brakes Lab

Step 1 of 3 · System Basics

20 pts

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STUDENT INFORMATION**Name:****Date:****LEARNING OUTCOMES**

- Identify disc brake components and configurations on the assigned vehicle
- Locate manufacturer specifications for rotor thickness, runout, and torque
- Describe the burnish-in procedure and the rotor matching process
- Identify common customer brake concerns and disc brake fade types

REQUIRED RESOURCES

- Assigned vehicle · ShopKey service info · Textbook · Computer
- Safety glasses and shop uniform · Hand tools · Brake lathe access

VEHICLE APPLICATION

Use your assigned vehicle. Default reference: 2014 Subaru Outback (any engine, 08/2014, w/ smart key).

Make	Model	Year	Engine
Transmission	Production Date	Model Code	Build Location

Research Questions

Use ShopKey, the textbook, and any provided training materials.

Note: Do not proceed to Step 2 until your instructor signs off this section.

- 1) Using ShopKey, identify the following torque specifications and rotor thickness for your assigned vehicle.

Brake Caliper Mounting Bracket Torque (ft-lb)	
Brake Caliper Slide/Guide Pin Torque (ft-lb)	
Rotor Minimum Thickness (mm or in)	

- 2) Using ShopKey, what is the minimum brake pad thickness for your assigned vehicle? Include units.

- 3) Using ShopKey and the textbook, describe the procedure for measuring rotor thickness variation (parallelism). Include the tool used and how many points around the rotor.

- 4) Using ShopKey and the textbook, list the appropriate grease for the caliper slide pins. Be specific about type.

- 5) Using ShopKey and the textbook, describe the optimal break-in procedure to burnish brake pads. Include speeds and number of stops.

- 6) Using ShopKey, what is the maximum allowable lateral runout for your assigned vehicle? Include units.

- 7) Using ShopKey and the textbook, explain the rotor matching (phase-matching) process. Why is it done and what are the steps?

- 8) Using ShopKey, search for TSBs related to the disc brake system. Document your search and findings.

SEARCH TERMS USED**TSBs FOUND - LIST NUMBER AND TITLE**

- 9) Using the textbook, identify and describe the types of disc brake fade. Include at least three types.

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- 10) Using ShopKey and the textbook, identify the main customer concerns related to disc brakes. List at least four with a brief description of each.

INSTRUCTOR SIGN-OFF - SYSTEM BASICS COMPLETE

Instructor Initials:

Date:

GRADE BREAKDOWN

QUESTION / SECTION	WHAT EARNS FULL CREDIT	POINTS
Q1 Specifications	All 3 specs filled correctly with units	3
Q2 Pad min thickness	Correct value with units from ShopKey	1
Q3 Thickness variation procedure	Tool, number of points, location described	2
Q4 Slide pin grease	Specific grease type identified (silicone synthetic)	1
Q5 Burnish procedure	Speeds, number of stops, cool-down described	2
Q6 Max runout	Correct value with units from ShopKey	1
Q7 Rotor matching	Why and how, with steps described	2
Q8 TSBs	Search terms documented AND TSBs listed (or 'none found' acceptable with terms)	2
Q9 Brake fade types	Three or more types identified with descriptions	2
Q10 Customer concerns	Four or more concerns with brief descriptions	2
Instructor sign-off	Instructor initialed the completed section	2
TOTAL - STEP 1: SYSTEM BASICS		20 pts

Reminder: Have your instructor initial above before uploading this PDF to Canvas.